

MHSA-TDL — Multi-Head Self-Attention Telemetry-Driven Deep Learning for Cluster Health Prediction and Failure Detection

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Abstract

Cloud platforms rely on continuous telemetry—CPU, memory, disk I/O, network, and scheduling—to keep clusters available, yet operators still need a fair comparison of attention-based monitors against hybrid and classical baselines on the same data and metrics. This study implements Multi-Head Self-Attention (MHSA-TDL) over multi-metric windows on a disclosed Google Cluster Trace (GCT) 2011 subset and scores Accuracy, Precision, Recall, Macro-F1, ROC-AUC, and prediction latency against Aldomi-style SelectKBest+GRU hybrids and classical RF/KNN/SVM monitors across five seeds. On the committed subset (N=12000 windows; fail-forced labelling), classical RF reaches Accuracy 0.944 ± 0.002 ; Aldomi SelectKBest+GRU-RF reaches Accuracy 0.943 ± 0.001 and Macro-F1 0.676; MHSA-Fused reaches Accuracy 0.922 ± 0.002 . MHSA is not claimed to outperform those monitors here. The fourth MHSA channel is sampled CPU because ClusterData 2011 has no network-byte column. Negative and near-parity outcomes are retained. The artefact supplies a reproducible GCT loader, baseline suite, and metric harness. Full GCT dumps, 2019 Borg cells, true network-byte channels, and live streaming remain beyond-CA2 residuals.

Introduction

Cloud computing underpins large-scale digital services through elastic, multi-tenant clusters. Telemetry streams—resource utilisation, I/O, and scheduling events—are the primary signal for predicting unhealthy machines or failing tasks before outages cascade. Research on Google Cluster Trace and related corpora has moved from threshold monitors to classical machine learning and hybrid deep models, for example feature selection plus GRU extractors with RF or KNN

heads as in Aldomi et al. (2026). Missed or late failure predictions drive SLA violations, wasted capacity, and costly firefighting. Attention mechanisms promise to capture multi-metric temporal dependencies without heavy manual feature engineering, but only comparative evaluation on a committed public trace and metric suite can show whether that promise holds. Beneficiaries are SRE and platform teams choosing monitors, and researchers benchmarking attention models against hybrid and classical baselines.

The formal research question asks what the performance of the proposed MHSA-TDL framework is in predicting cloud cluster health and detecting potential failures from cloud telemetry, compared with existing hybrid deep learning and traditional monitoring approaches. The work builds MHSA-TDL over multi-metric telemetry—CPU, memory, disk I/O, and network or scheduling channels as the schema allows—for cluster health and early failure detection, with success defined by trainable MHSA encoders in the artefact and a documented channel mapping. It trains and evaluates on Google Cluster Trace rather than synthetic-only data, requiring GCT-backed result tables with provenance such as `gct_load_meta.json` and `RESULTS_PROVENANCE`. Comparison against CA2-named baselines (Aldomi-style hybrid; RF/KNN/SVM/GRU-style monitors) uses the same split and seeds and shared metric columns for all methods. The formal metric suite—Accuracy, Precision, Recall, F1, ROC-AUC, and prediction latency, with confusion FN/FP analysis—is reported so that `results_summary.csv` and per-seed CSVs are populated for the GCT path.

The study follows a quantitative experimental design on a disclosed GCT 2011 subset (task events and usage join with fail-forced windows). Models include MHSA-Fused and MHSA-PerHead variants, Aldomi-style SelectKBest+GRU with RF or KNN heads, and classical RF/KNN/SVM plus a threshold baseline. Evaluation uses five seeds (42–46), macro metrics, fail-class metrics, and latency. Proxy-era Thapliyal underprediction narratives are non-binding relative to formal CA2.

Evaluation is offline on a subset, not a full multi-week 2011 dump or 2019 cells. Network bytes are unavailable in the 2011 schema, so sampled CPU substitutes. TensorFlow, Colab, and EC2 are training-compute options in CA2, not a requirement to deploy Lambda. Results may not generalise to other clusters or labelling policies. Window construction and FAIL labelling follow the artefact loader; class imbalance is handled as implemented in the harness; classical and hybrid baselines are

fair peers when trained on the same features and splits; and synthetic generators are for engineering tests only.

Section 2 reviews telemetry-driven failure prediction and the formal baseline family. Section 3 describes MHSA architectures and fusion choices. Section 4 covers the GCT loader, training harness, and baselines. Section 5 reports five-seed GCT results, including negative MHSA outcomes. Section 6 concludes and lists optional residuals outside the research floor.